safeguards as well. But what about transforming the surface environments not of asteroids or comets, but of planets? Could we live on Mars?

If we wanted to set up housekeeping on Mars, it's easy to see that, in principle at least, we could do it: There's abundant sunlight. There's plentiful water in the rocks and in underground and polar ice. The atmosphere is mostly carbon dioxide. It seems likely that in self-contained habitats—perhaps domed enclosures—we could grow crops, manufacture oxygen from water, recycle wastes.

At first we'd be dependent on commodities resupplied from Earth, but in time we'd manufacture more and more of them ourselves. We'd become increasingly self-sufficient. The domed enclosures, even if made of ordinary glass, would let in the visible sunlight and screen out the Sun's ultraviolet rays. With oxygen masks and protective garments—but nothing as bulky and cumbersome as a spacesuit—we could leave these enclosures to go exploring, or to build another domed village and farms.

It seems very evocative of the American pioneering experience, but with at least one major difference: In the early stages, large subsidies are essential. The technology required is too expensive for some poor family, like my grandparents a century ago, to pay their own passage to Mars. The early Martian poineers will be sent by governments and will have highly specialized skills. But in a generation or two, when children and grandchildren are born there —and especially when self-sufficiency is within reach—that will begin to change. Youngsters born on Mars will be given specialized training in the technology essential for survival in this new environment. The settlers will become less heroic and less exceptional. The full range of human strengths and deficiencies will begin to assert themselves. Gradually, precisely because of the difficulty of getting from Earth to Mars, a unique Martian culture will begin to emerge—distinct aspirations and fears tied to the environment they live in, distinct technologies, distinct social problems, distinct solutions—and, as has occurred in every similar circumstance throughout human history, a gradual sense of cultural and political estrangement from the mother world.